

PART II

SHORT-TIME MULTI-PHYSICS PHENOMENA IN REACTOR CORE

II-1 Prompt and Delayed Neutron Phenomena¹

In static reactor problems, the prompt and the delayed fission neutrons always appear together as the total number of fission neutrons. The fact that some of the fission neutrons are emitted as delayed neutrons has no consequence for static problems. The time dependence of the neutron flux, however, may be strongly influenced by the small fraction of the fission neutrons that is produced after time delays of about a second to several minutes. Even though this fraction is small, it may play a dominant role in many kinetics phenomena. Therefore, the production rates of delayed neutrons must be considered in detail.

In ^{235}U / ^{238}U -fueled thermal reactors at higher burnup, as well as in plutonium-fueled fast reactors, several isotopes contribute to the production of delayed neutrons. Thus, the delayed neutron data of several fissionable isotopes must be considered. A further complication appears in fast reactors where fissions are induced by neutrons in a wide energy range. Therefore, the isotope as well as the energy dependencies of the delayed neutron production need to be considered and are discussed in this chapter. Particular emphasis is placed on features of the delayed neutron data that have an impact on the theoretical formulation of the kinetics equations. The discussion of the physics of delayed neutron production is limited to a survey of the material necessary to understand the presented delayed neutron data.

Production of Prompt and Delayed Neutrons through Nuclear Fission

Nearly all of the neutrons produced as a result of a fission process are emitted "promptly," i.e., without noticeable delay. The prompt neutrons are emitted by the "direct" fission products immediately after the fission process since the excitation energy of the fission product nuclei is generally much larger than the separation energy for a neutron (i.e., neutron binding energy E_{Bn}). Figure II-1 shows schematically the nuclear levels of the two nuclei involved in the production of prompt neutrons. The difference in the total binding energies of the nuclides (A, Z) and $(A-1, Z)$ is merely the neutron binding energy E_{Bn} (note that $A = Z + N$). The excitation energy E^* in Fig. II-1 is $> E_{Bn}$. The typical decay time of such excited states may be 10^{-15} s or less, which is completely negligible.

¹ Sources:

1. Ott & Neuhold, *Introductory Nuclear Reactor Dynamics*, American Nuclear Society, 1985
2. Bedirhan Akdeniz, *Improved Neutron Kinetics for Coupled Three-Dimensional BWR Analysis*, PhD thesis, Pennsylvania State University, 2007

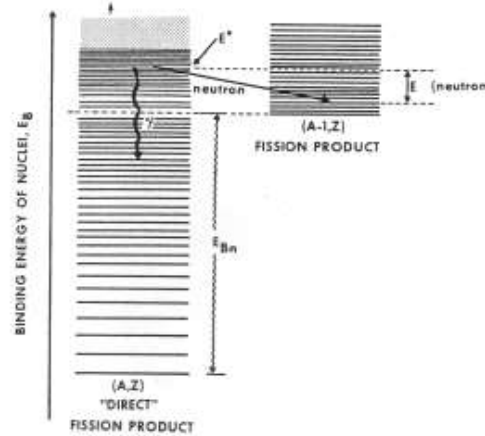


Fig. II-1 Schematic nuclear level structure leading to prompt neutron emission

Since E^* is larger than the neutron binding energy, there is competition between the neutron and gamma-ray emissions that leads to a strong straggling in the actual number of neutrons emitted from a fission event. Some fission product pairs may—with a certain probability—lose their excitation energy only by emission of gamma rays. In other cases, as many as six prompt neutrons may be emitted after a fission event [1]. The *average* number of prompt fission neutrons is denoted by ν_p .

After the fission process and the production of prompt neutrons is completed, the reaction products consist of

1. two radioactive nuclei
2. several prompt neutrons
3. several gamma rays.

None of the resultant fission product nuclei can directly emit an additional neutron. However, some of the fission product nuclei may decay into daughter nuclei for which the excitation energy is larger than the neutron binding energy; i.e., after a beta decay, the typical level scheme situation depicted in Fig. II-1 may in some cases occur again. Such nuclei may then immediately emit a neutron, which has been delayed by the comparatively long time it took such a nucleus to undergo a beta decay. Practically, the entire delay comes from the beta decay and only a negligibly small amount comes from the actual neutron emission [5].

The terms "precursor" nuclei and "emitter" nuclei are commonly used to denote the specific parent and daughter nuclei that produce delayed neutrons. Figure II-2 shows schematically the comparative nuclear level structure of the three nuclei involved in the production of delayed neutrons. The exceptional feature that allows the emission of a delayed neutron is that $E_{\beta}^{\max}$, the maximum electron energy in the beta decay, must be larger than the energy E_{Bn} required to separate a neutron, i.e.,

$$E_{\beta}^{\max} > E_{Bn} \quad (\text{II.1})$$

Some beta decays can then lead to excited states in the emitter nucleus above E_{Bn} . From such an excited state, the nucleus has a chance to emit a neutron, or alternatively emit a gamma quantum and assume temporarily an excited state below E_{Bn} . Consequently, not all of the possible "precursor" nuclei decays will lead to delayed neutron production. Since only those "parent nuclei" decays that really yield delayed neutrons are relevant for dynamics, "precursors" are defined as only that fraction of the parent nuclei that yields delayed neutrons.

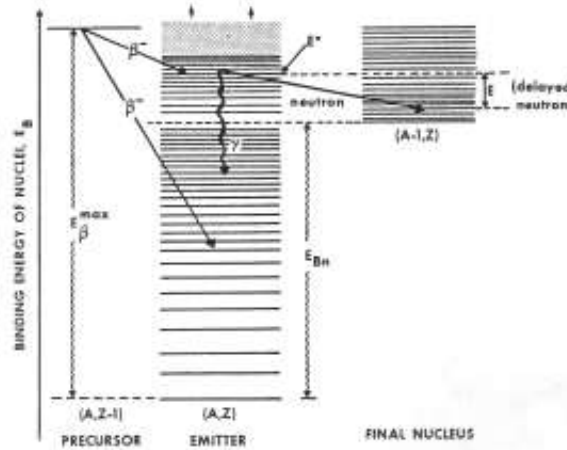


Fig. II-2 Schematic nuclear level structure leading to the emission of delayed neutrons.

It is thus obvious that the production of delayed neutrons after the precursor nucleus has decayed into the emitter nucleus is basically the same physical process as the production of prompt neutrons. However, the difference in the excitation energy and the neutron binding energy in delayed neutron emitter nuclei is normally much smaller than in the highly excited "direct" fission products. Therefore, delayed neutrons are emitted on the average with considerably smaller energies than prompt neutrons.

Total Delayed Neutron Yields

The average total number of neutrons, ν , which is composed of prompt and delayed neutrons, is expressed in terms of the relevant yields (neutrons per fission):

$$\nu = \nu_p + \nu_d \quad (\text{II.2})$$

The production of delayed neutrons was often described in the older literature by the delayed neutron "fraction,"²

$$\beta^{ph} = \frac{\nu_d}{\nu} \quad (\text{II.3})$$

The delayed neutron fraction, ν_d , appears then as a derived quantity, i.e., as the product of ν and β^{ph} :

$$\nu_d = \nu \beta^{ph} \quad (\text{II.4})$$

rather than directly as in Eq. (II.2). There are, however, two good reasons to favor the description in Eq. (II.2):

1. The yield ν_d is the more basic quantity and the use of the more basic quantity is preferred in general, unless there is an advantage to replacing by a less basic or artificially defined quantity.
2. Theoretical arguments [6] as well as newer measurements [7-9] show that ν_d in the range in which nearly all delayed neutrons are produced ($E \leq 4$ MeV) is practically independent of the energy of the fission inducing neutron, i.e.,

$$\nu_d(E) = \nu_d \approx \text{constant for } 0 \leq E \leq 4 \text{ MeV.}$$

The prompt yield ν_p (and thus ν , because of the small contribution of ν_d) shows, however, a significant energy variation in this range (an increase of $\sim 20\%$) [10]. Therefore, by using Eq. (II.3), an artificial energy dependence would be introduced into the description of delayed neutrons.

Figure II-3 shows [7, 8] schematically the energy dependence of ν_d for ^{235}U and ^{239}Pu . The lack of energy dependence below ~ 4 MeV is obvious. At higher energies, $\nu_d(E)$ decreases in general agreement with theoretical expectations [11] and recent measurements [7, 8]. This decrease is in strong disagreement with earlier measurements, which showed an increase by a factor of ~ 2 for all isotopes [12]. Since the fraction of delayed neutrons produced above 4 MeV is very small, ν_d is assumed independent of the incident neutron energy in the entire range of neutronics calculations.

The total delayed neutron yields, ν_{di} , are significantly different for different fissioned isotopes. There seem, however, to be two regularities [13] (see Table II-1):

1. The total delayed neutron yield increases with increasing atomic weight for a given element.
2. The total delayed neutron yield decreases with increasing number of protons.

² The somewhat clumsy notation $\beta^{ph} = \beta$ -physical is used to denote the ratio of Eq (2.3), the short notation, β is reserved for the frequently occurring β -effective, often denoted by β_{eff} .

Both regularities seem to be suggested by corresponding shifts in the abundances of fission products.

Table II-I gives the total delayed neutron yields for the most important uranium and plutonium isotopes. These values are used for the results presented in this chapter.

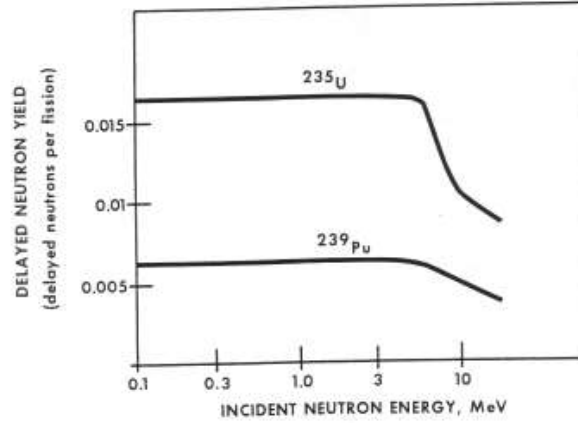


Fig. II-3 Energy dependence of the delayed neutron yields of ^{235}U and ^{239}Pu [7].

Table II-I: Total Delayed Neutron Yields (Delayed neutrons per fission)

Fission Nuclide	ν_d	Reference
^{233}U	0.0070 ± 0.0004	14
^{235}U	0.0165 ± 0.0005	14
^{238}U	0.0412 ± 0.0017	14
^{239}Pu	0.0063 ± 0.0003	14
^{240}Pu	0.0088 ± 0.0006	14
^{241}Pu	0.0154 ± 0.0015	15
^{242}Pu	0.0160 ± 0.0050	8

Yields of Delayed Neutron Groups

The "normal" situation is that the binding energy of the weakest bound neutron is larger, often much larger, than the maximum beta decay energy of the neighboring isotope. However, there are ~40 out of ~500 different fission product nuclides that have the special property required for being a delayed neutron emitter [Eq. (II.1)]. All 40 precursors have different lifetimes and therefore the corresponding neutrons will appear at different delay times. The consequence of the different precursor lifetimes is that the corresponding delayed neutrons will have a different effect on the time dependence of the neutron flux. Therefore, the different precursor lifetimes have to be accounted for in some way.

The most direct way to consider the effect of the differences in the precursor lifetimes is to take all of them into account individually. This approach has three serious drawbacks:

1. The lifetimes, and the abundancies in particular, of many precursors are not known accurately enough.

2. Even if the lifetimes and abundancies of all precursors were known accurately, their inclusion into the theoretical formulation of the kinetics problem would lead to an impractically lengthy set of differential equations.
3. Several precursors are themselves products of beta decays. The individual description of these precursors requires the inclusion of double or even multiple beta decay schemes. It has not been shown that this added complexity has any practical significance.

Since the use of the individual precursors in kinetics appears too complicated and still too inaccurate, an approach in which the delayed neutron data of all precursors are properly condensed needed to be found that allows a simple and yet sufficiently accurate theoretical description of reactor kinetics.

A suitable approach for obtaining condensed delayed neutron data was suggested historically by the fact that experimental knowledge on delayed neutron production was available primarily in the form of average source curves, $S_d(t)$. These source curves are obtained by exposing a sample of fissionable material to a very short neutron pulse, which instantaneously produces-through fission-a large number of precursors. The decay of these precursors results in the source of delayed neutrons $S_d(t)$ [14]. If n_f is the number of fissions that occurred in the sample during the flux pulse, then,

$$\nu_d n_f = \text{total number of precursors.}$$

The term $S_d(t)$ describes the decay rate of these precursors and thus the production rate of delayed neutrons. Figure II-4 shows a typical example of such a source curve, based on the data given in [14]. The decay rate of the single isotope ^{87}Br with its mean decay time of ~ 80 s is also shown. The experimental points in the actual experiment are average values of 80 measured decay curves. Physically, the decay curve $S_d(t)$ is a superposition of contributions from all precursors including those that are themselves beta decay products. It was shown in [14] that this complicated superposition $S_d(t)$ can be fairly accurately represented by just six exponential functions without including a buildup term, representing production of precursors through beta decay:

$$S_d(t) = n_f \sum_{k=1}^6 \nu_{dk} \lambda_k \exp(-\lambda_k t) \quad (\text{II.5})$$

Equation (II.5) represents a delayed neutron source that results from the decay of six "average" groups³ of precursors, all of them produced at $t = 0$. The "delayed group" yields ν_{dki} and the average "delayed group" decay constants λ_{ki} are obtained for each isotope i by a least-squares fit of the right side of Eq. (II.5) to an experimentally determined left side:

³ The notation "group" may often lead to confusion, especially when different energy groups are used in the same equation or sentence. Therefore, we use consistently the notation "delay group" and reserve the notation "group" for energy group. To avoid this kind of confusion, some authors use the shorter term "family" for "delay group" introduced in Ref. 16.

$$S_{di}(t) = n_{fi} \sum_{k=1}^6 \nu_{dki} \lambda_{ki} \exp(-\lambda_{ki} t) \quad (\text{II.6})$$

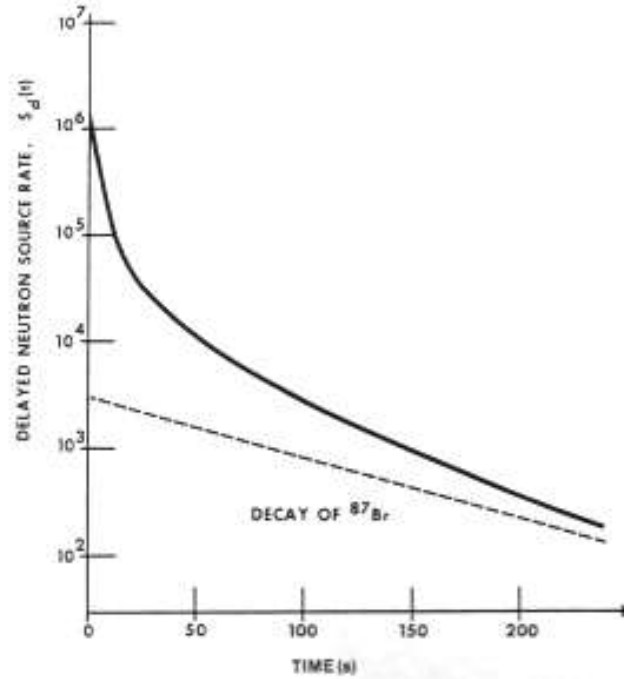


Fig. II-4 Delayed neutron source following a fission pulse [14].

This six delay group structure is in general use in reactor kinetics. For a complete presentation of the data as well as comparisons with the results of other authors (see [1]).

As long as ^{235}U is the dominant fuel in a thermal reactor, practically all of the delayed neutrons are produced from fission products of ^{235}U . Then the delayed neutron sources $S_d(t)$ consist of six terms as in Eq. (II.6). In high-burnup light water reactor fuel and in a fast breeder reactor (FBR), however, there are several nuclides that contribute to the delayed neutron source. The ^{238}U contribution in an FBR is comparable to the ^{239}Pu contribution because its delayed neutron yield is about seven times larger than that for ^{239}Pu , thereby largely compensating for its lower fission rate. Other contributions such as those of ^{240}Pu and ^{241}Pu may also be of practical significance. The contribution of ^{242}Pu is generally small. In any case, if more than one isotope is of importance, the simplicity of the condensed representation, Eq. (II.5), is lost. The total delayed neutron source has to be found as a sum of all isotopic contributions; e.g., in the case of a neutron pulse:

$$S_d(t) = \sum_i n_{fi} \sum_{k=1}^6 \nu_{dki} \lambda_{ki} \exp(-\lambda_{ki} t) \quad (\text{II.7})$$

Fortunately, the isotope dependence of the decay constants is not very pronounced. Most of them differ only within their statistical errors, which become apparent from Table II-II in which the delay group decay constants of ^{239}Pu are compared with those of ^{235}U , ^{238}U and ^{240}Pu . Therefore, the use of a single set of isotope-independent decay

constants, λ_k , can be expected to fit the experimental delayed neutron sources with only an insignificant loss of accuracy compared to Eq. (II.6), i.e.,

$$S_{di}(t) = n_{fi} \sum_{k=1}^6 \nu_{dki} \lambda_k \exp(-\lambda_k t) \quad (\text{II.8})$$

The results of such a fit [17] are given in Table II-III. The delay yields in Table II-III are used for the calculations presented in this book. The great advantage of having a single set of decay constants is that the precursor concentrations and the delayed neutron sources can be readily summed up for all isotopes, and macroscopic cross sections can be introduced in the same way as that for prompt fission neutrons (see Sec. 3-2A). Although the yields of Table II-III have been suggested for use in fast reactors, they are also applicable to thermal reactor problems. For thermal reactor problems, one would prefer the use of the λ_k values of ^{235}U instead of ^{239}Pu [17]. However, the differences in the λ 's are so small that they cannot affect the accuracy of the resulting fit.

Table II-II: Comparison of Precursor Decay Constants in Six Delay Groups for Various Isotopes* (s^{-1})

Delay Group	^{239}Pu	^{240}Pu
1	0.0129 ± 0.0002	0.0129 ± 0.0004
2	0.0311 ± 0.0005	0.0313 ± 0.0005
3	0.134 ± 0.003	0.135 ± 0.011
4	0.331 ± 0.012	0.333 ± 0.031
5	1.26 ± 0.12	1.36 ± 0.21
6	3.21 ± 0.26	4.04 ± 0.78
	^{238}U	^{235}U
1	0.0132 ± 0.0003	0.0127 ± 0.0002
2	0.0321 ± 0.0006	0.0317 ± 0.0008
3	0.139 ± 0.005	0.115 ± 0.003
4	0.358 ± 0.014	0.311 ± 0.008
5	1.41 ± 0.07	1.40 ± 0.081
6	4.02 ± 0.21	3.87 ± 0.37

*From [14].

Instead of the consistent fit represented by Eq. (II.8), other more or less inconsistent eliminations of the isotopic dependence of the decay constants are in widespread use.

The longer living precursors that contribute significantly to delayed neutron production are probably all known (delay groups 1 to 3). Delay group yields may therefore also be defined by the composition of individual precursor yields within a proper lifetime bracket [4]. Some neutron emission probabilities and precursor production abundancies are known experimentally. Most of them, however, have to be estimated theoretically so that composite yields can be calculated. In [4], results for delay groups 1 to 4 of the isotopes ^{235}U , ^{238}U , and ^{239}Pu are presented. The values for groups 1 and 2 agree for all isotopes with the values of Table II-III within their error limits (delay group 1 is assumed to consist of ^{87}Br only, and for delay group 2 there are only three major contributors).

For delay groups 3 and 4, the agreement is still remarkably good even though most of the values are outside of the error limits of Table II-III. Full agreement cannot be expected because of the inherent difference in the definitions of the composite yields [4] and the yields resulting from a least-squares fit of a source curve [14].

Table II-III: Delayed Neutron Group Yields Consistently Fit to a Single Set of Decay Constants*

Delay Group	$\nu_{dk}, \frac{\text{neutrons}}{\text{fission}} \times 100$	
	^{233}U	^{235}U
1	0.053 ± 0.003	0.060 ± 0.005
2	0.197 ± 0.012	0.364 ± 0.013
3	0.175 ± 0.025	0.349 ± 0.024
4	0.212 ± 0.013	0.628 ± 0.015
5	0.047 ± 0.014	0.179 ± 0.014
6	0.016 ± 0.006	0.070 ± 0.005
	^{238}U	^{232}Th
1	0.049 ± 0.005	0.143 ± 0.012
2	0.540 ± 0.027	0.776 ± 0.046
3	0.681 ± 0.092	0.843 ± 0.099
4	1.526 ± 0.096	2.156 ± 0.103
5	0.836 ± 0.033	0.838 ± 0.053
6	0.488 ± 0.036	0.204 ± 0.034
	^{239}Pu	^{240}Pu
1	0.024 ± 0.002	0.028 ± 0.003
2	0.176 ± 0.009	0.237 ± 0.016
3	0.136 ± 0.014	0.162 ± 0.046
4	0.207 ± 0.012	0.314 ± 0.038
5	0.065 ± 0.007	0.106 ± 0.026
6	0.022 ± 0.003	0.039 ± 0.008
	^{241}Pu	^{242}Pu
1	0.019 ± 0.004	0.036 ± 0.013
2	0.369 ± 0.010	0.263 ± 0.086
3	0.276 ± 0.045	0.270 ± 0.075
4	0.534 ± 0.089	0.607 ± 0.177
5	0.310 ± 0.043	0.279 ± 0.055
6	0.032 ± 0.007	0.145 ± 0.059

*From [17]. The new delayed neutron yields are to be used with the following set of isotope-independent decay constants (s^{-1}):

$$\lambda_1 = 0.0129 \pm 0.0002 \quad \lambda_2 = 0.0311 \pm 0.0005$$

$$\lambda_3 = 0.134 \pm 0.003 \quad \lambda_4 = 0.331 \pm 0.012$$

$$\lambda_5 = 1.26 \pm 0.12 \quad \lambda_6 = 3.21 \pm 0.26.$$

Emission Spectra of Delayed Neutrons

The excitation energies of delayed neutron emitters are much smaller than the excitation energies of prompt neutron emitters. Whereas the prompt fission neutrons have an average energy near 2 MeV and their spectrum extends even beyond 10 MeV, the delayed neutrons are emitted with a much smaller average energy and thus with a softer spectrum than prompt neutrons.

Since different delay groups are composed of different precursors with different energy ranges ($E^* - E_{Bn}$; see Fig. II-2), the emission spectra depend on the delay group; they are denoted by $\chi_{dk}(E)$.

If the energy dependence of the delayed neutron emission is included in Eq. (II.5), the following energy-dependent delayed neutron source results from a fission pulse for a single isotope:

$$S_d(E, t) = n_f \sum_k \chi_{dk}(E) \nu_{dk} \lambda_k \exp(-\lambda_k t) \quad (\text{II.9})$$

Let us consider three extreme cases:

1. the "initial" emission spectrum:

$$S_d(E, 0) \propto \sum_k \chi_{dk}(E) \nu_{dk} \lambda_k \quad (\text{II.10})$$

2. the stationary emission spectrum, which results from integrating Eq. (II.9) over all times:

$$S_d^{stat}(E) \propto \sum_k \chi_{dk}(E) \nu_{dk} \quad (\text{II.11})$$

3. the asymptotic emission spectrum, i.e., the case in which the longest living group provides the overriding contribution:

$$S_d^{as}(E) \propto \chi_{dk}(E) \text{ with } k = 1 \quad (\text{II.12})$$

The comparison of these extreme cases shows that the composition of the total delayed neutron emission spectrum may vary strongly with time and the static spectrum may have a different composition compared to any time-dependent one.

For many years, the only emission spectra for delayed neutron groups were those measured by Batchelor and McK. Hyder [18] for the first four delay groups of ^{235}U fissioned by thermal neutrons. More recently, Fieg [19] measured delayed neutron emission spectra for ^{238}U , ^{239}Pu , and also ^{235}U , with fission initiated by 14-MeV neutrons. The yields at 14 MeV are significantly lower than those in the dominating fission range (see Fig. II-3). This suggests possible delayed neutron energy spectrum differences between the fission range and 14 MeV. Fortunately, the dependencies on isotopes and on the initiating neutron energy appear to be relatively small, as indicated by the average energies presented in Table II-IV, which shows that the average energies for

each delay group are about the same for all three isotopes and for thermal and 14-MeV fission (i.e., all error bands overlap). This independence is not surprising—delay group 1 consists only of the precursor ^{87}Br in all cases. All other groups consist of contributions of several precursors. A variation in the emission spectra and, thus, in the average energies of the other delay group neutrons is due to small variations in the relative abundance of different precursor isotopes. Also, the variation of the average energy with decreasing lifetime of the precursor groups appears to be very minimal, hardly outside of the experimental errors (10% for [19] and 2 to 8% for [18]). Therefore, the common practice of using the emission spectrum measured for delay group 4 for delay groups 5 and 6 also seems to be reasonable. However, the emission spectra should be extended down to low energies, [20] i.e., below the experimental lower limit of ~ 100 keV in Ref. 18. A set of such χ_d values in a half lethargy group structure is presented in Table 2-V along with the total neutron emission spectrum [21]. Evaluations [9] for ENDF/B-IV recommend the use of the data of Ref. 19 for all delay groups except delay group 2. Recently reported measurements [22] are recommended for delay group 2.

Table II-IV: Average Energies for Various Delay Groups, Isotopes, and Fissioning Energies

Isotope	Average Energies in Delay Groups (keV)					Reference
	1	2	3	4	5	
^{235}U (thermal)	250 ± 20	460 ± 10	405 ± 20	450 ± 20	-	18
^{235}U (thermal)	277 ± 28	484 ± 48	447 ± 45	432 ± 43	-	19
^{235}U (14MeV)	286 ± 29	458 ± 46	432 ± 43	480 ± 48	-	19
^{238}U (14MeV)	278 ± 28	468 ± 47	443 ± 44	425 ± 43	382 ± 40	19
^{239}Pu (14MeV)	296 ± 30	481 ± 48	411 ± 41	430 ± 43	-	19

Evaluation of Basic Delayed Neutron Data

There has been a significant progress in developing nuclear data libraries (evaluated nuclear data files) for the last 25 years. Different updates and versions of the US Evaluated Nuclear Data Files (ENDF/B), the Joint European Files (JEF) and the Japanese JENDL libraries have been issued and subsequently validated and utilized. There have been also attempts to improve the basic delayed neutron data as part of these libraries to be consistent with the rest of cross-section data in the libraries. This is important since later in the process of evaluating delayed neutron parameters with a lattice physics code this cross-section data is utilized in the homogenization and collapsing process. In addition, at this stage different correction techniques can be utilized in the evaluation process of delayed neutron fraction to account for the delayed neutron data, in order to obtain an accurate estimate of the effective delayed neutron fractions.

This section reviews the status of delayed neutron data (later, the current methods for the evaluation of delayed neutron fractions will be discussed and summarized with the study of the choice of the best correction methods to be utilized in transient applications). The performed studies on the evaluation of basic delayed neutron data are discussed and

summarized with the current status of the nuclear data libraries. This section concludes with recommendation and proposal for the library to be utilized in point kinetics and coupled three-dimensional (3-D) kinetics/thermal-hydraulic calculations.

Delayed neutron data uncertainties may result in undesirable conservatism in the design and operation of nuclear power plants. Interest in improving the accuracy of the delayed neutron (DN) data began to grow in early 1990s. As a result an international working group was formed at the Nuclear Energy Agency (NEA), OECD - NEA WPEC SG6 - to review, improve and validate DN data [23]. The SG6 has reviewed the DN data at three different levels: individual precursor (**microscopic**) level; aggregate precursors (**macroscopic**) level, in-pile (**integral**) measurement level. This group completed its work at 1999 with the following major results:

- Recommending a new set of DN yields for the major fissile isotopes.
- Developing a new 8-group model that has a better physical basis than the presently used 6-group model.
- Investigating the need that the DN group parameters be made an explicit function of the incident neutron energy.

As a result of the NEA WPEC SG6 work conclusions concerning the DN data for major actinides were summarized and published in a special issue of *Progress of Nuclear Energy* on DN data of 2002 [24]. The recommended data is compared with the DN data currently in use. The Tuttle's 1975 and 1979 evaluations of total yields, together with Keepin's 6 group parameters representing time dependence, are still widely used and comparison between these, the data in the current nuclear data libraries (ENDF/B, JEF and JENDL), and the recommended one by SG6 are very important.

SG6 Studies

In summary, the studies performed for the delayed neutron data have addressed the following issues:

- the programs of beta-effective measurements;
- the energy dependence of delayed neutron yields;
- evaluations of total delayed neutron yields;
- time-dependent data;
- the energy spectra of delayed neutrons.

At the microscopic level, the measurements and associated evaluations have significantly improved the database. These measurements and evaluations are being incorporated in the fission product yield and radioactive decay data files in the major nuclear data libraries such as ENDF/B, JEF, and JENDL. Although the accuracy of the data on microscopic level has improved, for β_{eff} (effective delayed neutron fraction) calculations, the reliance must be still placed on the macroscopic measurements of the DN emission data for major isotopes and validation of the data using integral reactor measurements of β_{eff} and time-dependent effects.

There is a tendency for the measurements made in the different experimental programs to yield different values of β_{eff} . The uncertainties associated with measurements are estimated and the adoption of an average of the adjusted values based on the different studies is suggested. It is considered that using these averaged values an accuracy of $\pm 3\%$ (1 s.d.) will be achieved in β_{eff} calculations for conventional thermal reactors and $\pm 2\%$ to $\pm 3\%$ for fast reactors fueled with the major actinide isotopes. The papers presented in the above-mentioned special issue by Okajima et al (2002) and Fort et al (2002) show that the agreement between the measured and calculated β_{eff} values is good, within $\pm 3\%$ for the data in the current nuclear data libraries, ENDF/B-VI, JEF-2.2, and JENDL-3.2. However, ENDF/B-VI yield data underestimate β_{eff} values measured in the MOX fueled cores, which have a high dependence on ^{238}U . Further inter-comparisons of techniques, and measurements made on cores, would be helpful in understanding the differences and give confidence in the measured values.

The incident neutron energy dependence of total delayed neutron yields is believed to be small - the difference between thermal reactor and fast reactor spectrum averaged values is at most few percent. Fission product precursor summation calculations give much larger differences and the reason for these needs to be understood. The uncertainty about the possible variations through resonances also prevents a clear conclusion being reached about the relationship between thermal and fast spectrum yields. More work is needed to define the energy dependence of total yields. However, very accurate relative measurements ($\pm 1\%$) will be required for the major actinide isotopes. For the secondary isotopes, the uncertainties are larger and more measurements having a lower precision would be useful. Accurate relative measurements of the energy dependence of total yields, and of the fractional yields used to represent time dependence, would enable the systematic of the interrelationships to be explored in more detail. The dependence on incident neutron energy may be further studied but a capability to treat it can be incorporated in advance.

Improved representations of the time dependence of the delayed neutron emission was proposed by Spriggs, Campbell, and Pisalkin (1999) based on a new 8-group precursor structure for delayed neutron group constants and spectra. The new structure is defined based on current knowledge of the half-lives of the dominant precursor isotopes. In particular, the half-lives of the first three groups have been fixed at the half-lives of the three longest-lived dominant precursors. The new 8-group structure is characterized by the same set of half-lives for all fissioning isotopes and for fission induced by neutrons of different energies. Therefore, data in the new structure can be correctly used in reactor kinetics calculations by solving only nine differential equations (eight for the precursors in different groups and one for the neutron density). On the contrary, the Keepin's 6-group structure (characterized by different sets of half-lives for different isotopes and for different incident neutron energies) in principle requires the solution of six differential equations for each fissioning isotope and for each different incident neutron energy.

The use of the 8 group representation of time dependence proposed by Spriggs, Campbell and Piksaikin, with the half-lives of the three predominant long-lived precursors being explicitly treated, is recommended as having a better physical basis than the traditional 6 group representation of Keepin. The relative yields and energy spectra derived by them are also recommended.

The NEA WPEC SG6 conclusion paper summarizes the recommended total yield values in [24] as:

<u>U-235 thermal</u>	<u>U-235 fast</u>	<u>U-238 fast</u>	<u>Pu-239 thermal</u>	<u>Pu-239 fast</u>
<u>0.0162</u>	<u>0.0163</u>	<u>0.0465</u>	<u>0.00650</u>	<u>0.00651</u>

Based on these spectrum-averaged values energy dependent values have been derived, suitable for inclusion in nuclear data library files. The data is given in [24].

Status of Nuclear Data Libraries

Delayed neutron parameters using the ENDF/B-VI basic nuclear data:

- The initial DN data prepared on the basis of the first release of ENDF/B-VI (1989) suffered somewhat from the “parallel” way of data evaluation typical for the environment of ENDF/B versions’ development.
- The initial set of ENDF/B-VI 6-group DN parameters have been found inadequate in many applications.
- In 1993 – an improved set of DN data based on ENDF/B-VI was released.
- In 2002 – the latest update of the DN parameters based on ENDF/B-VI was published (W. Wilson and T. England, “Delayed Neutron Study Using ENDF/B-VI Basic Nuclear Data”, Progress in Nuclear Energy, Vol. 41, No. 1-4, pp. 71-107, 2002).
- The above study has confirmed that there is no compelling reason for reactor design codes to depart from using six DN groups especially if they are used with variable lambdas.
- The use of 8 groups with variable lambdas is better than 6 groups, but the difference is not significant in terms of the uncertainty in DN fraction. However, there is a penalty for using 8 groups with variable lambdas.

In JEFF-3.0 for some important fissile nuclides the DN data is presented in 8 groups with energy spectra depending on incident neutron energy. The code developers of core dynamics models should adopt a new scheme to utilize such data:

- Using a mixture of 8 and 6 group data.
- Modeling the energy spectra dependence on incident neutron energy i.e. using different delayed neutron groups for the different energy group.

The question is under what circumstances would code developers adopt such scheme and would the industry be willing to go that way.

Review Questions

1. What are delayed neutron precursors?

2. Give the typical level scheme of precursor-emitter-daughter nuclei that leads to the production of delayed neutrons.
3. What is the known or estimated approximate number of possible precursors?
4. Name the quantities used to describe the physics of precursors and delayed neutrons.
5. Give approximately the total precursor yields of ^{235}U , ^{238}U , and ^{239}Pu .
6. Sketch the typical E dependence of $\nu_d(E)$ for ^{235}U and ^{239}Pu .
7. What is the typical difference in the emission spectra of prompt and delayed neutrons? Give typical values for the respective average emission energies.
8. How many delayed neutron groups (families) are generally used per fissioning isotope?
9. What is the disadvantage of using isotope-dependent decay constants?
10. Describe briefly the experiment for which delayed neutron data are derived.
11. Describe briefly the way in which delayed neutron data are obtained from the experimental results.
12. Give the approximate mean lifetime of the slowest and the fastest decaying precursor group.

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